CS:5980
Foundations of Embedded Systems

Dynamical Systems

Part I

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Dynamical Systems

Controller interacting with the physical world via sensors and actuators

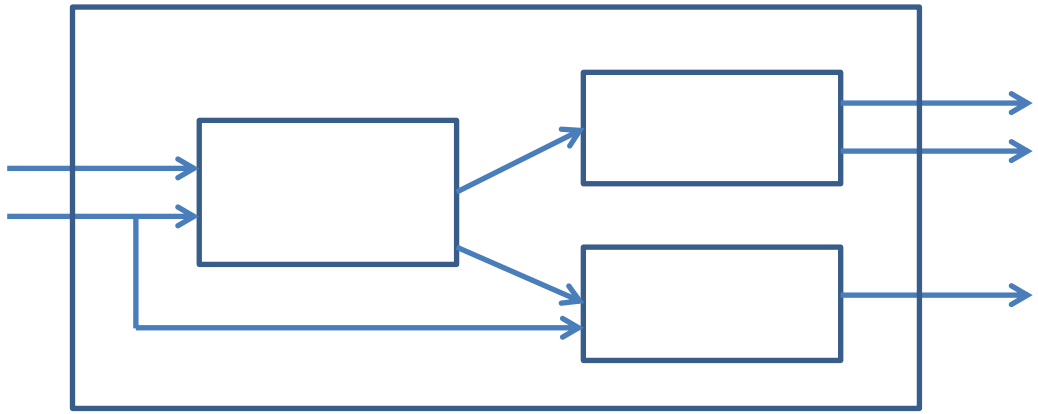
- Thermostat controlling temperature
- Cruise controller regulating speed of a car

System variables: (measures of) physical quantities evolving continuously over time

- Temperature, pressure, velocity …

Continuous-time models using differential equations

Model-Based Design



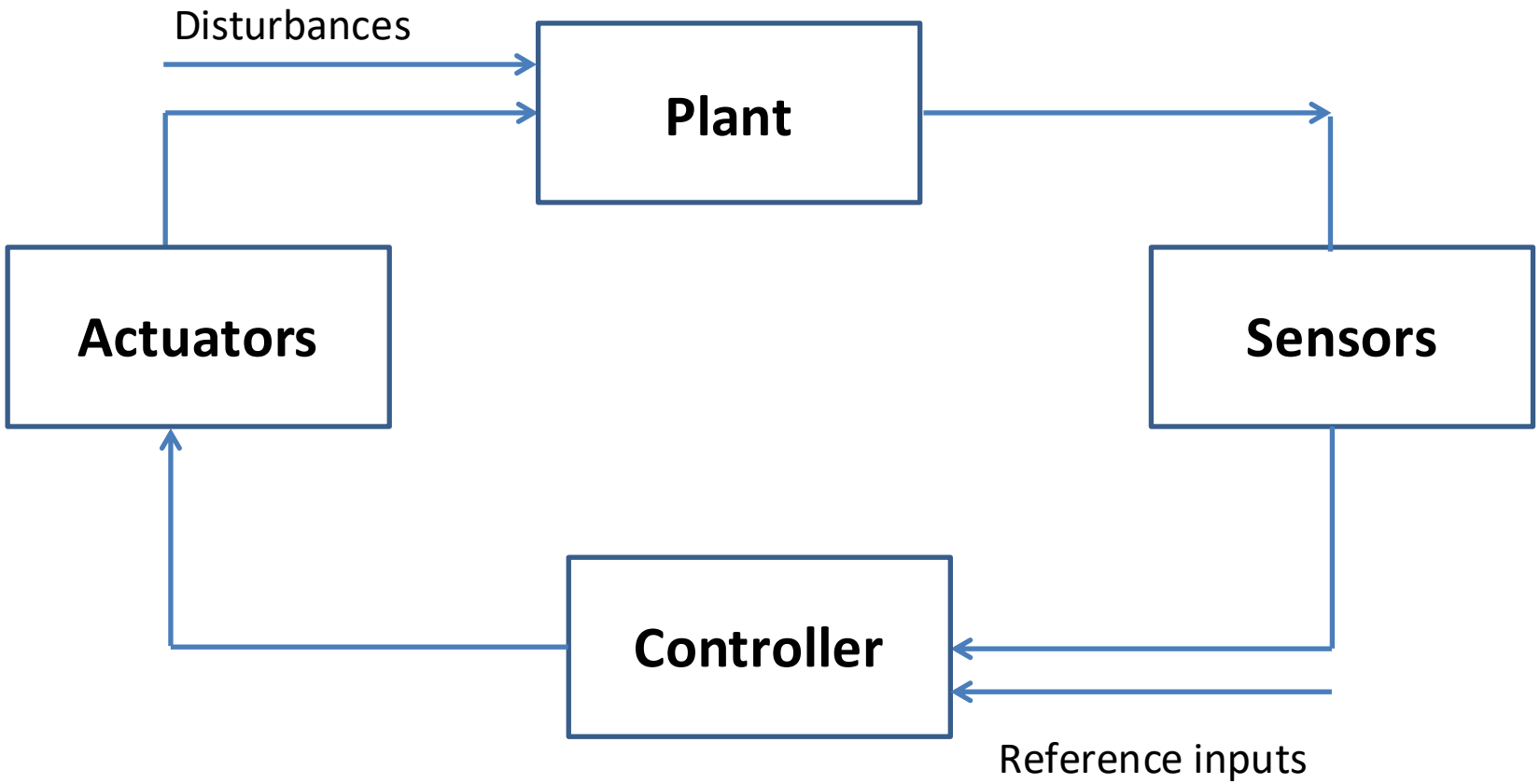
Block Diagrams

- Widely used in industrial design
- Tools: Simulink, Modelica, RationalRose…

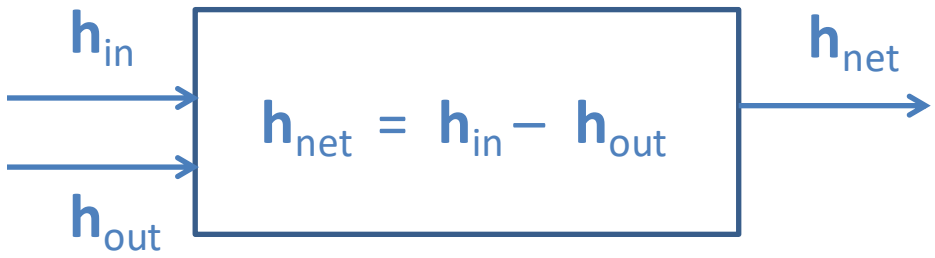
Key question: what is the execution semantics?

- Similar to synchronous model but continuous-time instead of discrete-time

Traditional Feedback Control Loop



Example: Heat Flow



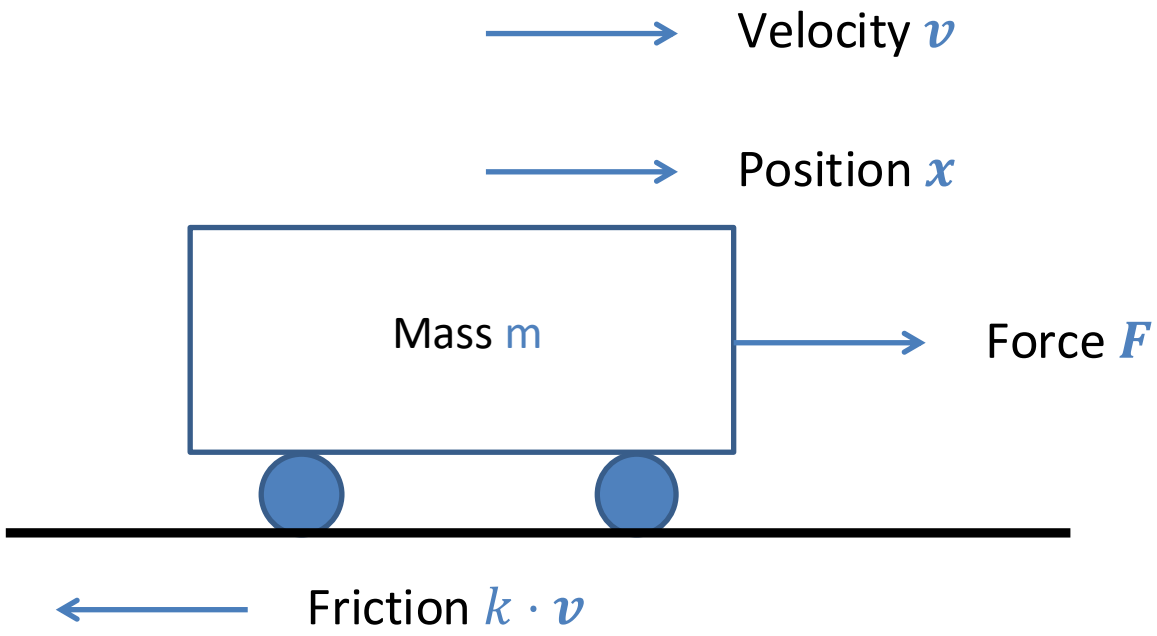
Input variables: h_{in} and h_{out} of type **real**

Output variable: h_{net} of type **real**

State variables: none

- *Signal*: assignment of values to variables as function of time t
- At each time t, value of output signal $h_{net}(t)$ equals $h_{in}(t) - h_{out}(t)$
- Output as a function of inputs/state, specified using algebraic equations (as opposed to assignments)

Car Model



- v, x and F are all functions of time t; k is a friction constant
- Newton's law of motion gives:

$$F - k \cdot v = m \cdot d^2x/dt$$

Notation

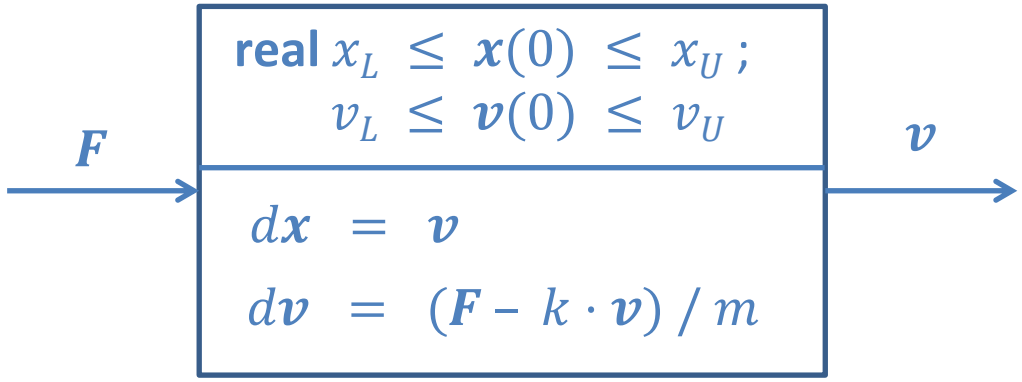
First derivative of function $f(t)$ with respect to t :

- $df(t)/dt$ (full notation)
- df/dt (with the understanding that f is a function of t)
- df (dimension t is implicit)
- f' (same as df)
  - $\dot{f}$ (same as df)

Second derivative of function $f(t)$ with respect to t :

- $d^2f(t)/dt^2$ (full notation)
- d^2f/dt^2 (with the understanding that f is a function of t)
- d^2f (dimension t is implicit)
- f'' (same as d^2f)
  - $\ddot{f}$ (same as d^2f)

Continuous-time Car Component



- The value of output variables is defined in terms of input and state variables
- For each state variable $\boldsymbol{s}$, its *rate of change* $d\boldsymbol{s}/dt$ is defined in terms of input and state variables

Executions of Car Component

Input signal: function $\boldsymbol{F}: \mathbb{R}_{\geq 0} \to \mathbb{R}$ that gives value of force over time

- should be continuous or piecewise-continuous

Given an initial state (x_0, v_0) and input signal $\boldsymbol{F}$, the execution of the system is defined by state-signals (ie., functions)

$$\boldsymbol{x}: \mathbb{R}_{\geq 0} \to \mathbb{R} \quad \text{and} \quad \boldsymbol{v}: \mathbb{R}_{\geq 0} \to \mathbb{R}$$

that satisfy the initial-value problem:

$$\boldsymbol{x}(0) = x_0 \qquad d\boldsymbol{x}(t)/dt = \boldsymbol{v}(t)$$

$$\boldsymbol{v}(0) = v_0 \qquad d\boldsymbol{v}(t)/dt = d^2\boldsymbol{x}(t)/dt^2 = (\boldsymbol{F}(t) - k \cdot \boldsymbol{v}(t))/m$$

Executions of Car: Example 1

Suppose force $\boldsymbol{F}(t)$ is always 0, and initial position is 0.

We need to solve:

$$\begin{cases} \boldsymbol{x}(0) = 0 \\ \boldsymbol{v}(0) = v_0 \\ d\boldsymbol{x}/dt = \boldsymbol{v} \\ d\boldsymbol{v}/dt = k \cdot \boldsymbol{v}/m \end{cases}$$

Solution:

- Velocity decreases exponentially fast, converging to 0

$$\boldsymbol{v}(t) = v_0 \cdot e^{-k \cdot t/m}$$

- Position converges exponentially fast to $m \cdot v_0/k$

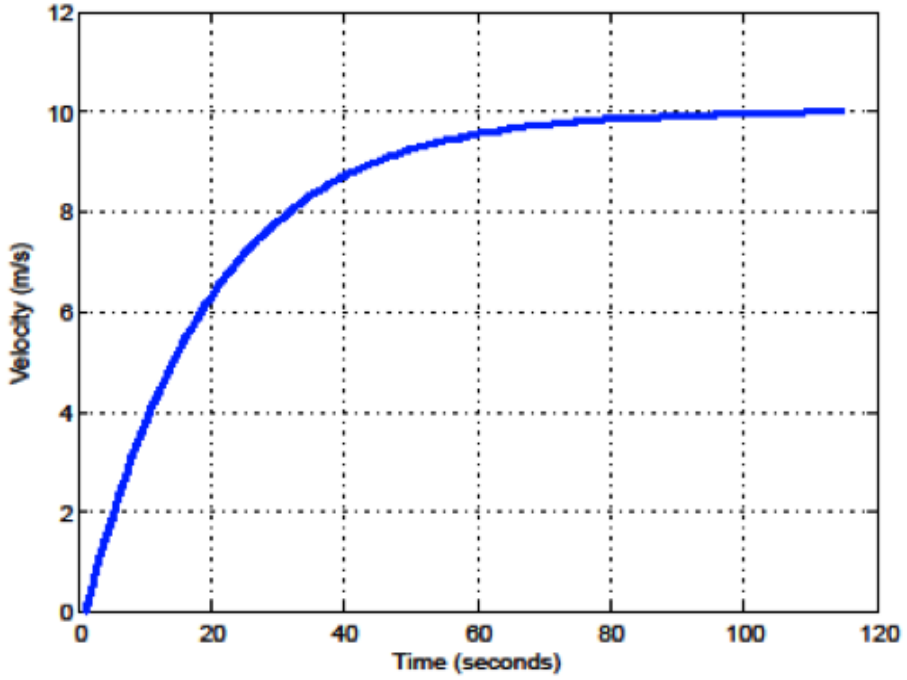
$$\boldsymbol{x}(t) = (m \cdot v_0/k)(1 - e^{-k \cdot t/m})$$

Executions of Car: Example 2

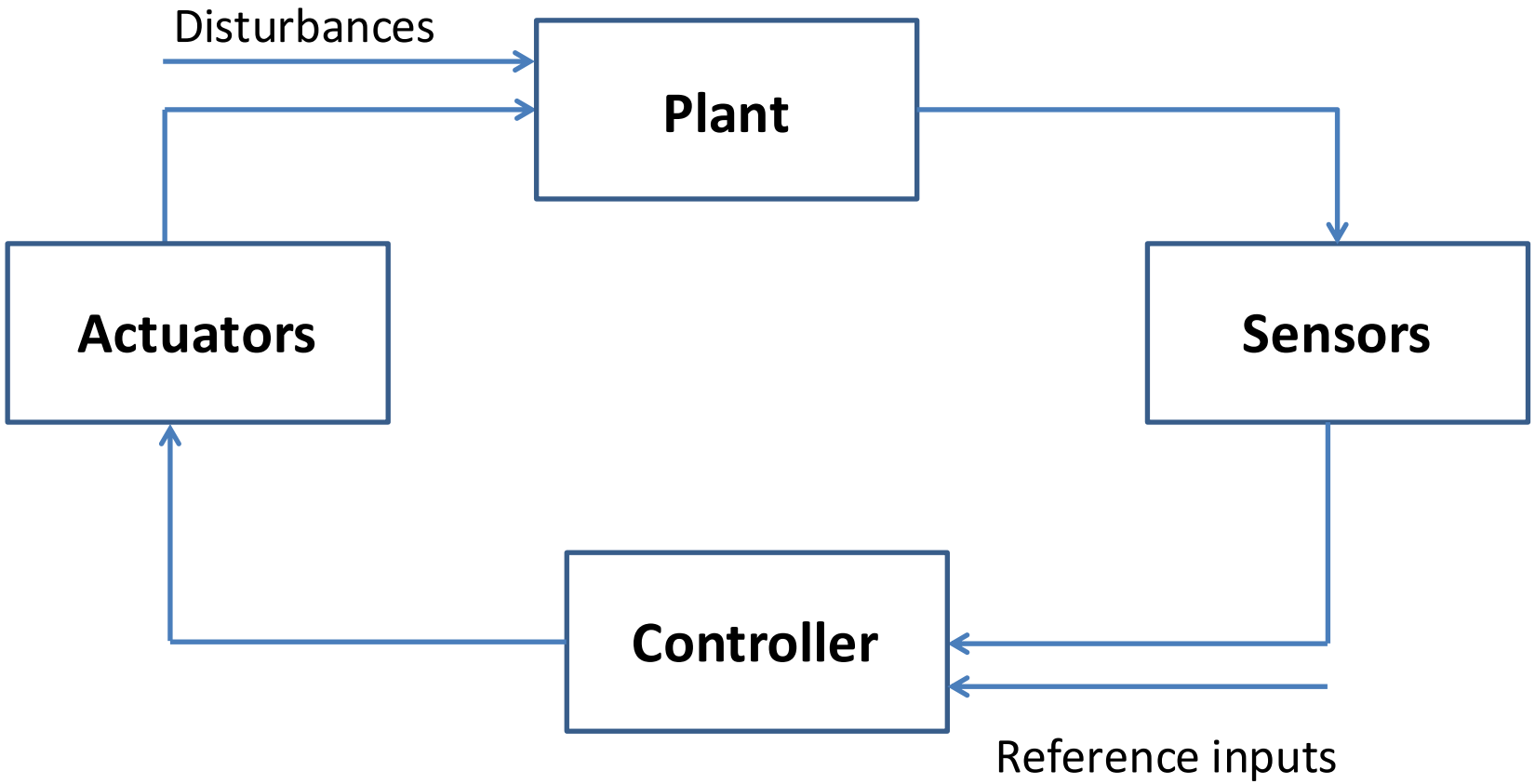
- Suppose initial position is 0, initial velocity is 0, and force is constant F_0. Then, to get executions, we need to solve:
  - $\boldsymbol{x}(0) = 0$
  - $\boldsymbol{v}(0) = 0$
  - $d\boldsymbol{x}/dt = \boldsymbol{v}$
  - $d\boldsymbol{v}/dt = (F_0 - k \cdot \boldsymbol{v})/mc$

Compute the solution using MATLAB

- Mass $m_c = 1000kg$
- Coefficient of friction $k = 50$
- Force $F_0 = 500\ Newton$
- Velocity converges to $10m/s$



Traditional Feedback Control Loop



Continuous-Time Component Definition

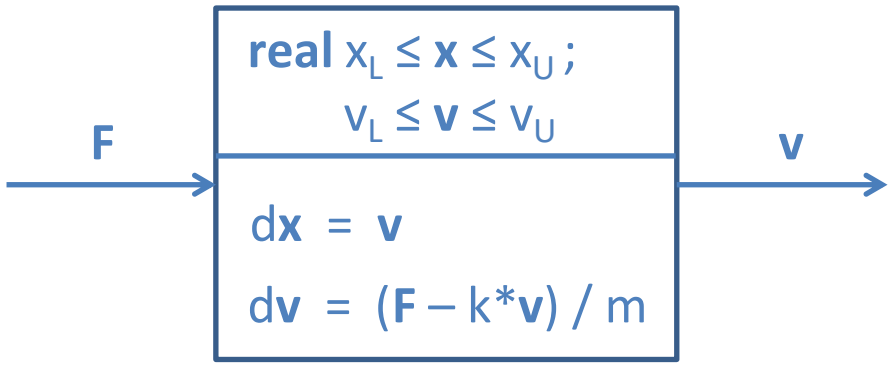
- Set I of real-valued *input variables*; type is either **real** or real interval **real**[L, U]
- Set O of real-valued *output variables*
- Set S of real-valued *state variables*
- *Initialization* Init specifying set $[\![\text{Init}]\!]$ of initial states
- For each output var **y**, a real-valued expression h_y over I ∪ S
- For each state variable **x**, a real-valued expression f_x over I ∪ S

Execution

Given an input signal **I**(t) : $\text{real}_{>=0} \rightarrow \text{real}^{|\mathbf{I}|}$, an *execution* consists of a differentiable state signal **S**(t) and output signal **O**(t) such that

1. **S**(0) is in $[\![\text{Init}]\!]$
2. For each output var **y** and time t, $\mathbf{y}(t) = h_y(\mathbf{I}(t), \mathbf{S}(t))$
3. For each state var **x**, $d\mathbf{x}(t)/dt = f_x(\mathbf{I}(t), \mathbf{S}(t))$

Continuous-time Component Car



1. The value of output variables is defined in terms of input and state variables
2. For each state variable **s**, its *rate of change* ds is defined in terms of input and state variables

Existence and Uniqueness

Given an input signal **I**(t), when are we guaranteed the system has at least/exactly one execution?

- The input signal should be (at least piecewise) continuous but answer also depends on right-hand sides of equations defining state and output dynamics

This related to classical theory of Ordinary Differential Equations (ODEs)

Consider the initial value problem

$$d\mathbf{x}/dt = F(\mathbf{x}) \; ; \quad \mathbf{x}(0) = \mathbf{x}_0 \, , \qquad \mathbf{x}(t) \text{ is n-dimensional vector}$$

- When do we have a unique differentiable function as a solution for **x**?

Solution Existence

***Initial value problem*:**

$$d\mathbf{x}/dt = F(\mathbf{x}) \;;\; \mathbf{x}(0) = \mathbf{x}_0 , \qquad \mathbf{x}(t) \text{ is } n\text{-dimensional vector}$$

The problem has a solution $\mathbf{x}(t)$ if function F is continuous

Example: of when solution does not exist:

$$dx/dt = \text{if } (x = 0) \text{ then } 1 \text{ else } 0$$

It is natural to require all right-hand-side expressions h_y and f_x in the definition of a continuous-time component to be continuous

- Will we see the discontinuous case in Chap. 9 (Hybrid Systems)

Continuous Function

Definition of continuity relies on a given notion of distance $\|\quad\|$ between points (e.g., Euclidean distance)

A function $f: \mathbb{R}^m \rightarrow \mathbb{R}^n$ is *(uniformly) continuous* if

for all $\varepsilon > 0$,

there is a $\delta > 0$ such that

for all $u, v \in \mathbb{R}^m$,

if $\|u - v\| < \delta$ then $\|f(u) - f(v)\| < \varepsilon$

Solution Uniqueness

Initial value problem:

$$d\mathbf{x}/dt = G(\mathbf{x}) \;;\; \mathbf{x}(0) = \mathbf{x}_0, \quad \mathbf{x}(t) \text{ is n-dimensional vector}$$

Theorem: There exists a unique solution $\mathbf{x}(t)$ if the function G is Lipschitz-continuous

Examples:

- A linear function such as $(\mathbf{F} - k\,\mathbf{v}) / m$ is Lip-continuous
- Quadratic function x^2 is Lip-continuous if domain of x is bounded

Counterexamples:

- $x^{1/3}$ is not Lipschitz-continuous: $dx/dt = x^{1/3}$; $x(0) = 0$ has multiple solutions:
  1. $\mathbf{x}(t) = 0$
  2. $\mathbf{x}(t) = (2t/3)^{3/2}$

Lipschitz-Continuous Function

Informally, a function f Lipschitz-continuous if
there is a constant upper bound on how much f's output changes

A function $f: \mathbb{R}^m \to \mathbb{R}^n$ is *Lipschitz-continuous* if there exists a constant c such that

for all $u, v \in \mathbb{R}^m$, $\quad \|f(u) - f(v)\| \leq c \cdot \|u - v\|$

Lipschitz-Continuous Component

Definition: A continuous-time component has *Lipschitz-continuous dynamics* if

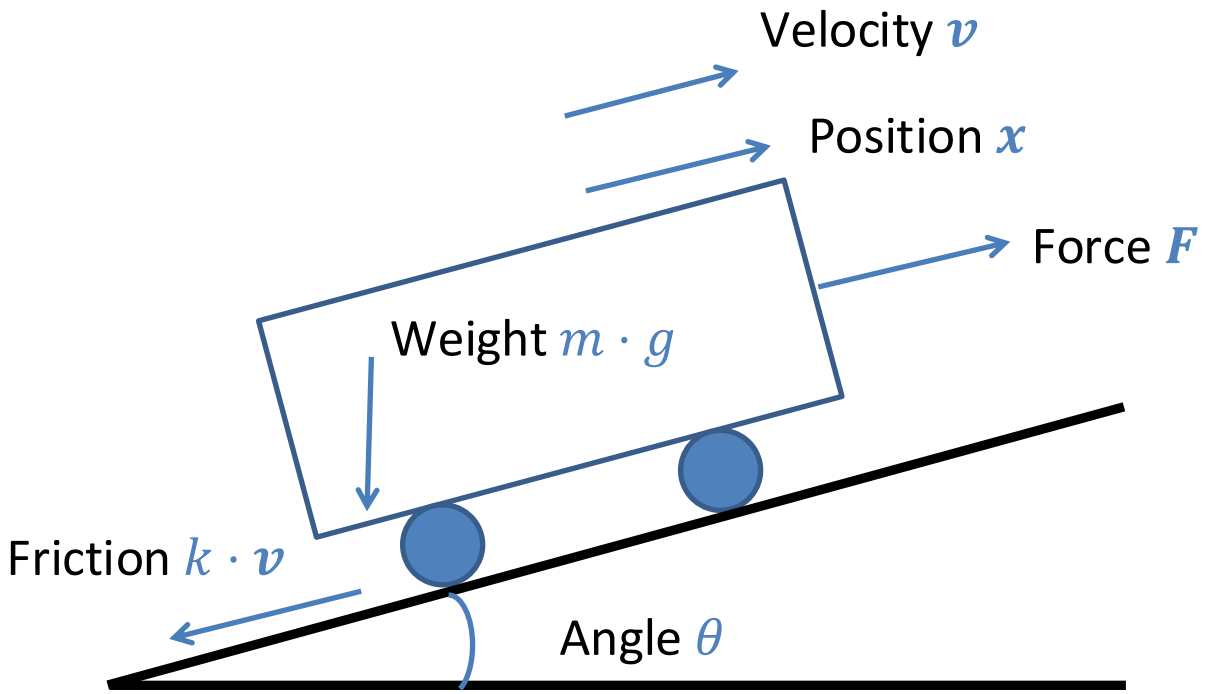
- each expression h_y corresponding to output variable **y** is a Lipschitz-continuous function of $I \cup S$
- each expression f_x corresponding to state variable **x** is a Lipschitz-continuous function over $I \cup S$

Theorem: Given a continuous input signal **I**(t), a component with Lipschitz-continuous dynamics has unique and continuous response signals **S**(t) and **O**(t)

Note: Continuity of output signals means they can be fed to other components in a block diagram

Henceforth, we will consider only Lipschitz-continuous components

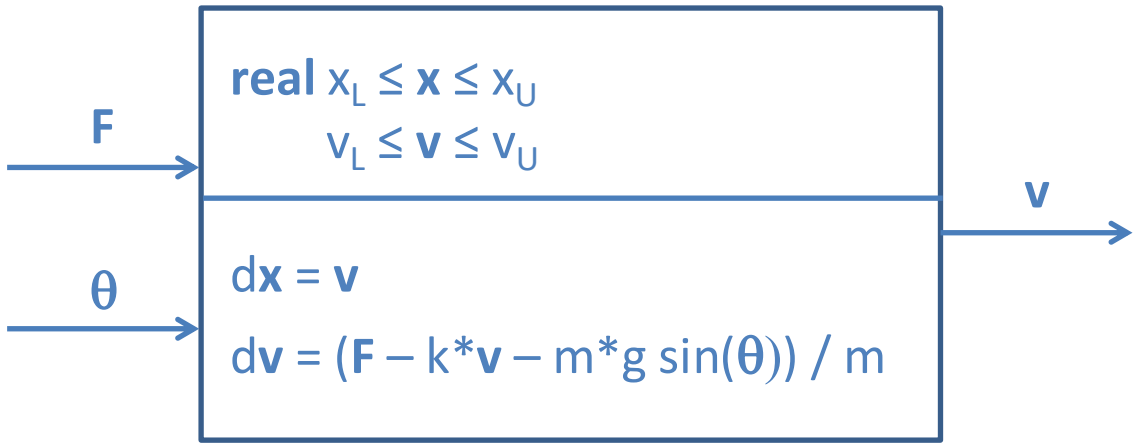
Car on a non-level road



Newton's law of motion gives

$$F - k \cdot v - m \cdot g \sin(\theta) = m \cdot d^2x/dt$$

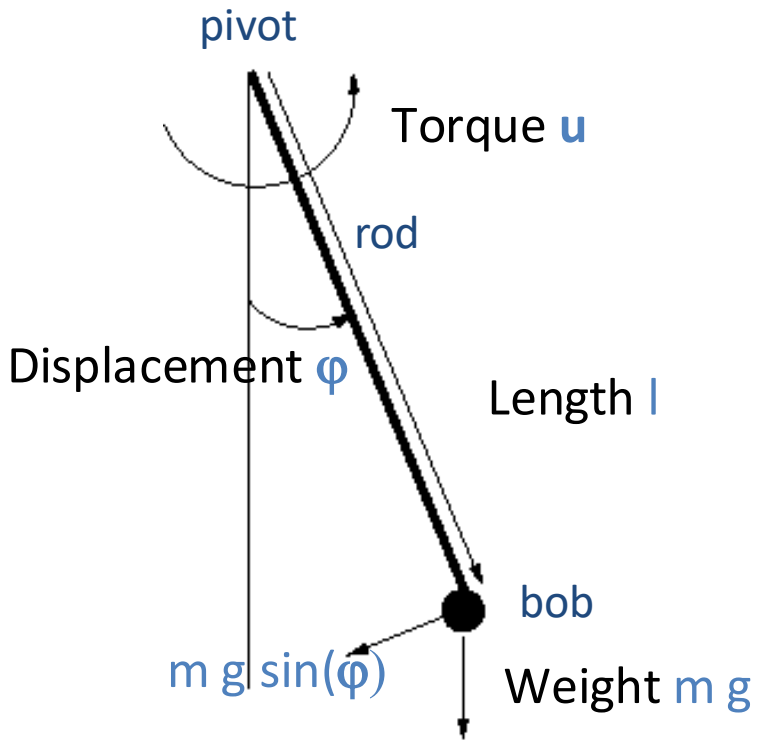
Continuous-time Car Component Revised



The road's slope, denoted by θ, models *disturbance*, or an uncontrolled input

Design problem: Find a controller with input **v** and output **F** such that the composed system works correctly for all continuous input signals $\theta(t)$, with $\theta(t)$ always in $[-\pi/6, \pi/6)$

Simple Pendulum



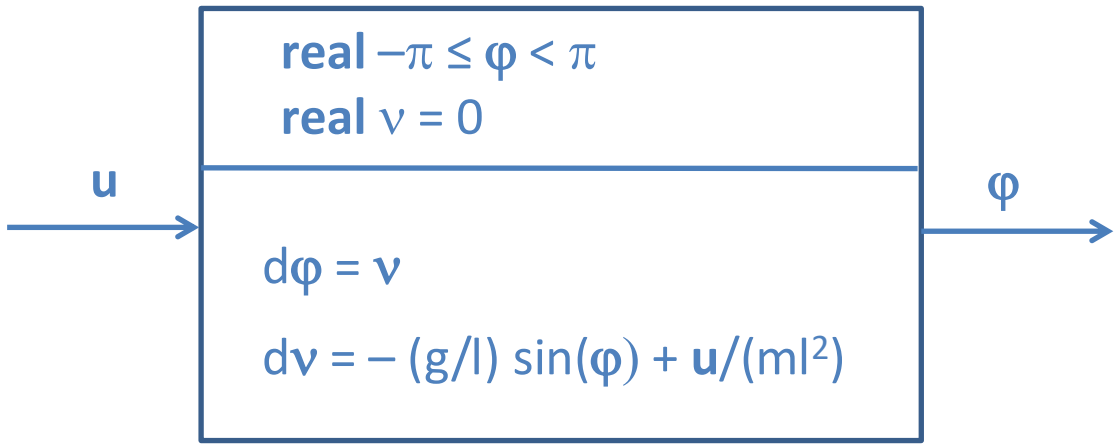
Assumptions: frictionless pivot, massless rod, massive bob

Dynamics captured by second-order non-linear differential equation:

$$m\,l^2\,(d^2\varphi/dt^2) = \mathbf{u} - m\,g\,l\,\sin(\varphi)$$

Pendulum Model

$$m\, l^2\, (d^2\varphi/dt^2) = \mathbf{u} - m\, g\, l \sin(\varphi)$$



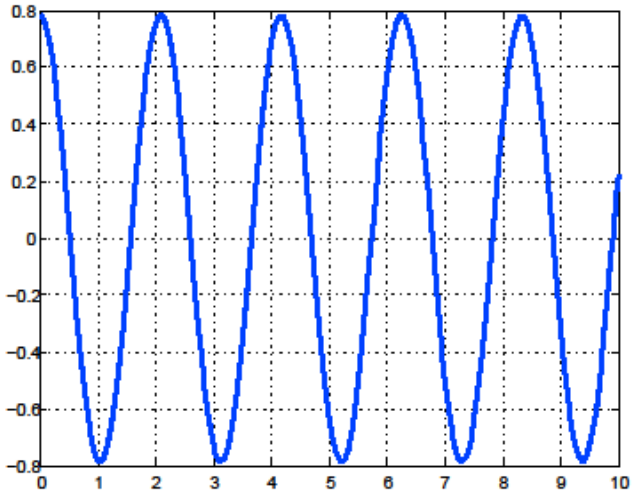
u: torque

φ: angular displacement (in radians)

ν: angular velocity

Angular Displacement

- External torque = 0
  Initial displacement = $\pi/4$
- Oscillatory motion plotted by MATLAB
- What are the equilibria of this pendulum?



Credits

Notes based on Chapter 6 of

Principles of Cyber-Physical Systems
by Rajeev Alur
MIT Press, 2015